

SIDDHANT S. ALARWAD

Python Developer | AI & LLM Application Engineer

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PROFESSIONAL SUMMARY

Python Developer with 1+ year of professional experience building and deploying production applications, now focused on applied AI. Built an offline LLM assistant on Ollama (Llama 3.2) using prompt engineering, intent classification, model guardrails and streaming inference, and a computer-vision pipeline (OpenCV, PyTorch) that reconstructs 3D models from video. At work, delivers Python automation and real-time data-acquisition software running on a factory floor. Strong in Python, OOP, multi-threading, SQLite and MySQL, with working knowledge of Docker, Git, Linux and Google Cloud Platform. Anthropic-certified in AI fluency, agent skills and Vertex AI. Immediate joiner, based in Bangalore.

TECHNICAL SKILLS

AI & LLM: LLM Integration (Ollama, Llama 3.2), Prompt Engineering, Intent Classification, Model Guardrails, Streaming Inference, Context Management, Computer Vision, AI/ML Fundamentals

AI Libraries: PyTorch, OpenCV, NumPy, ONNX Runtime, Open3D

Programming: Python, SQL, Bash

Core Engineering: OOP, Multi-threading (QThread), Concurrency, Modular Architecture, Error Handling, Unit Testing (pytest)

Application Development: PyQt5, PySide6, CLI Tools, PyInstaller (standalone Windows builds)

Data & Databases: MySQL, SQLite, Schema Design, Data Pipelines, openpyxl (Excel automation), SMTP automation

Cloud & DevOps: Google Cloud Platform (Compute Engine, Cloud Run, Cloud Storage, VPC, IAM), Docker, Terraform, GitHub Actions, CI/CD, Git, Linux

Industrial / IIoT: Siemens S7 (snap7), Modbus TCP, PLC Integration, SCADA

Practices & Tools: Agile / SDLC, Code Review, Debugging, Production Support, VS Code, PyCharm

AI & SOFTWARE PROJECTS

Haven — Offline LLM Assistant with Safety Guardrails

Tech: Python, PySide6, Ollama (Llama 3.2), SQLite, Prompt Engineering

- Built a fully offline chat application (~4,900 lines) that runs a local language model through Ollama and stores conversations in SQLite, so no user data leaves the device.
- Engineered a three-layer prompting system around a 3B-parameter model — persona constraints, per-message intent classification and conversation-state tracking — with output validation that automatically regenerates non-compliant replies.
- Tuned inference for responsiveness: sentence-boundary stream buffering (~90% fewer UI repaints), a startup warm-up with keep-alive that removed a 5–15 second cold start, and context trimming to bound per-request cost.

MeshForge — Video-to-3D Reconstruction & 3D-Printing Studio

Tech: Python, PyQt5, OpenCV, PyTorch, Open3D, NumPy, pytest

- Built a desktop and CLI tool (~5,300 lines, 44 modules) that turns a video or photos of an object into a watertight 3D model exported to STL, 3MF, OBJ, PLY and GLB.
- Designed a pipeline with three reconstruction strategies (visual-hull carving, structure-from-motion, monocular depth) and automatic fallback between them.
- Integrated deep-learning models — MiDaS/DPT monocular depth in PyTorch and ONNX Runtime segmentation — with classical computer-vision pose estimation (SIFT, RANSAC); covered by 50+ pytest tests.

PROFESSIONAL EXPERIENCE

Python Developer | SonyTech, Bangalore

July 2025 – Present

Develop, test and deploy Python applications for industrial and business automation — real-time data acquisition, database-driven reporting and email automation — and support them at client sites.

Multi-PLC Real-Time Data Acquisition System (SCADA) — Foundry Sand Plant

Tech: Python, PyQt5, snap7 (Siemens S7), MySQL, Multi-threading, PyInstaller

- Built a real-time desktop application that polls two Siemens PLCs concurrently over Ethernet and monitors 25+ process parameters, replacing 100% of manual shop-floor data logging.
- Engineered fault-tolerant, multi-threaded polling that separates network failures from data-address errors and auto-reconnects every 5 seconds, keeping monitoring running unattended.
- Designed dual-mode MySQL logging (10-second telemetry plus PLC-triggered batch records) and a password-protected runtime configuration panel.

Automated Production Reporting & Email Dispatch Service

Tech: Python, QThread, MySQL, SQLite, openpyxl, SMTP, PyInstaller

- Built a background service that queries daily production data, generates formatted Excel reports and emails stakeholders automatically, retiring a manual daily reporting task.
- Wrote shift-aware, index-friendly SQL with cross-midnight boundaries so overnight output is reported against the correct production day.
- Added TLS-secured SMTP dispatch with attachments and a 30-second health check on the database and mail server.

Delivery & Support

- Packaged applications as standalone Windows executables with PyInstaller and deployed them to client machines with no Python runtime required.
- Provided production support: diagnosed live issues from logs, fixed defects and released updates based on user feedback.

EDUCATION

Master of Computer Applications (MCA) | VTU, Belagavi | 2024 | CGPA 8.0 / 10

Bachelor of Computer Applications (BCA) | Gogte College of Commerce, Belagavi | 2022

CERTIFICATIONS

- Anthropic (2026): Claude with Google Vertex AI; AI Fluency; Agent Skills; Subagents; Claude 101
- Google: Foundations of Cybersecurity (Coursera) | Cisco: Introduction to Cyber Security, Cyber Security Essentials

ADDITIONAL INFORMATION

Availability: Immediate joiner | **Languages:** English, Hindi, Kannada, Marathi